

Elemental Experiments in Resolution Geometry

Chambers, Recurrence Classes, Admissible Deformations, and the Failure of the Early CTMT Sufficiency Claims

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September 27, 2026

Abstract

We replace the earlier CTMT elemental computation by a protocol-explicit Resolution Geometry (RG) formulation. The old construction inserted chemical groups into its regularizer, imposed lanthanide endpoint anchors and monotonicity, accepted mass-like modulation from phase sensitivity, and interpreted scalar second differences as rupture. A deterministic hostile battery shows that these steps do not characterize elemental organization: nonzero sensitivity need not identify the deformation, Tikhonov stability can reconstruct the wrong continuation, scalar rupture is not protocol invariant, and groupwise penalties can manufacture recovery. What survives is a conditional characterization. A local elemental chamber is a maximal connected region of constant transported admissible RG type away from a declared discriminant. An elemental group is not one stratum; it is a recurrence class of compatible local chambers across periods. The characterization locks mathematically relative to a declared experiment and transport protocol. Empirical identification of the actual chemical groups remains open until measured multichannel elemental data are processed without group labels entering the observables, covariance, kernel, transport, threshold, or discriminant selection.

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1 Position in the RG theorem ladder

The observation map and its kernel pair force only observational equivalence, the quotient, fibre saturation, universal factorization, and the kernel groupoid. Fisher geometry is a separate imported metric module; the resolved projector is characterized relative to a declared comparison metric and threshold; transport and gluing require identity and cocycle closure; a canonical connection from (J, Σ, K) remains open. Consequently no chemical partition, physical scale, or elemental transport may be smuggled into the forced root.

2 Declared elemental experiment

Definition 1 (Elemental admissible experiment). *For each atomic number Z and control state u , declare*

$$\mathfrak{E}_{Z,u} = (\mathcal{E}_{Z,u}, \Gamma_{Z,u}), \quad \Gamma = (G, \tau, N, S, C, T),$$

where $\mathcal{E}$ is the complete statistical experiment and Γ contains the comparison metric, resolution threshold, nuisance convention, stability and conditioning gates, coarse-graining rules, and transport data. Its characterized RG object is

$$\text{RG}_\tau(\mathfrak{E}_{Z,u}) = (Q_{Z,u}, g_{F,Z,u}, P_{Z,u}, W_{Z,u}, T, \Gamma_{Z,u}),$$

with $P_{Z,u} = \mathbf{1}_{(\tau,\infty)}(G^{-1}F)$ and $W_{Z,u} \subseteq \text{im } P_{Z,u}$ after all admissibility gates.

3 Early claims and their corrected status

3.1 AAA

The early criterion $\partial_a P \neq 0$ establishes sensitivity only. The attack uses $y = \theta + a$, for which $\partial_a y = 1$ but the Fisher matrix has rank one for two parameters. Thus a is not separately identifiable. With an independent channel $y_2 = \theta - a$, the Fisher spectrum becomes $(2, 2)$ and the direction resolves.

Proposition 1 (Admissible deformation gate). *A deformation a is RG-admissible only if its tangent direction survives quotienting, nuisance closure, the resolved projector, conditioning, stability, coarse-graining, and transport gates. Phase sensitivity is neither sufficient for identifiability nor for admissibility.*

3.2 Inversion and regularization

A Tikhonov estimator selects a member of a declared variational class. The battery constructs a piecewise latent field and an underdetermined observation map for which strong smoothing yields a low residual but large latent error. Regularization success therefore does not prove physical continuation, chamber membership, or elemental structure.

3.3 Rupture

A robust second difference is a useful candidate detector, not an invariant boundary certificate. The battery inserts a pure protocol-scale jump into a smooth signal; the raw rupture localizes at the scale change and is suppressed after normalization. A genuine chamber boundary requires a stable change of RG type, such as projector rank, generalized spectral multiplicity, nuisance rank, or transport class, with a positive discriminant margin.

3.4 Groups and lanthanides

The old objective partitioned Z into named chemical groups and imposed groupwise smoothness; the lanthanide interval additionally received endpoints and monotonicity. Those computations test compatibility with the supplied partition. They do not discover or characterize the partition. Lanthanide contraction can become evidence for a long chamber only after group and endpoint ablation, independent observables, and transport-stability tests.

4 Chambers and recurrence

Definition 2 (Transported type). *The local type $\mathcal{I}_Z$ is the collection of declared invariants preserved by an admissible comparison map, including resolved rank, generalized spectral data, nuisance type, conditioning and stability class, and any protocol-approved quotient invariants.*

Definition 3 (Elemental chamber). *An elemental chamber is a maximal connected interval in the chosen control base on which $\mathcal{I}_Z$ is locally constant and neighboring comparison maps are admissible.*

Definition 4 (Elemental group). *An elemental group is an equivalence class of chambers in distinct periods whose transported RG types are isomorphic, with identity, inverse, and cocycle closure within declared uncertainty.*

Thus a group is not a connected stratum in atomic-number space. It is a recurrence class of local strata.

Theorem 1 (Conditional elemental recurrence characterization). *Let $\{\mathfrak{E}_Z\}$ be a compatible family of regular finite-dimensional admissible experiments. Assume: (i) positive separation from all declared discriminants; (ii) locally constant resolved and nuisance ranks inside each candidate chamber; (iii) admissible transports preserving the declared type; (iv) identity, inverse, and cocycle closure; and (v) stability under admissible perturbation and sensor ablation. Then maximal connected constant-type regions are chambers, and isomorphism under the closed transport system defines recurrence classes of chambers. These recurrence classes are unique relative to the declared experiment, protocol, and transport data.*

Proof. Off the discriminant, regularity gives local constancy of the discrete invariants. Maximal connected components therefore define chambers. Identity, inverse, and cocycle laws make transported isomorphism reflexive, symmetric, and transitive, hence an equivalence relation on chambers. Its equivalence classes are the recurrence classes. Uniqueness is relative, because G, τ , nuisance conventions, stability rules, and transport are protocol data rather than consequences of the quotient alone. $\square$

5 Hostile battery

The deterministic script uses seed 20260927 and registers nine checks. All nine pass. The checks establish: sensitivity without identifiability; recovery after an independent channel; nuisance confounding; wrong but stable Tikhonov continuation; non-invariant scalar rupture; manufactured group recovery under label leakage; recovery of repeated chamber types only after declared transport; failure under cocycle defect; and a stable projector-rank chamber boundary with positive spectral margin.

Early statement	Verdict
Phase sensitivity makes AAA admissible	Rejected as sufficient; retained as a candidate gate.
Tikhonov success reveals elemental structure	Rejected.
Second-difference rupture is a structural boundary	Rejected as invariant; retained as a candidate detector.
Groupwise reconstruction discovers groups	Rejected as circular.
Lanthanide anchors prove a long stratum	Rejected without ablation and independent channels.
Groups are connected strata	Rejected.
Groups are recurrence classes of compatible chambers	Locked as a conditional characterization.
Actual chemical groups satisfy that characterization	Open empirical claim.

6 Empirical lock protocol

A final empirical test must use measured multichannel observations with no chemical group labels in J , Σ , K , feature selection, threshold selection, transport estimation, or discriminant detection. It must pre-register nuisance conventions and uncertainty, discover local chambers, test transport and cocycle closure, form cross-period recurrence classes, and only then compare the recovered classes with chemical groups. Negative controls must include channel ablation, label permutation, protocol rescaling, alternative admissible metrics, threshold sweeps, and held-out periods.

Non-claim 1 (No empirical periodic-table proof). *The present battery is a logical and synthetic characterization attack. It proves neither that the actual chemical groups are RG recurrence classes nor that a preferred elemental experiment is determined without physical input.*

Non-claim 2 (No CTMT ontology). *No phase kernel, universal mass modulation, background causal cone, or CTMT-specific ontology is required. The result belongs to the declared-experiment and stratified-transport layers of RG.*

7 Conclusion

The early elemental computation is replaced rather than extended. AAA becomes an admissible-deformation problem; inversion becomes estimator selection under an explicit protocol; rupture becomes a candidate discriminant diagnostic; local periodic positions become chambers; and chemical groups become prospective recurrence classes of compatible chambers. The mathematical characterization locks conditionally. The physical identification remains an explicit experiment.

Reproducibility

Companion artifacts: `rg_elemental_final_attack.py`, `rg_elemental_final_attack.json`. Deterministic seed: 20260927.